

Indian mathematics

Indian mathematics emerged in the [Indian subcontinent](#)^[1] from 1200 BCE^[2] until the end of the 18th century. In the classical period of Indian mathematics (400 CE to 1200 CE), important contributions were made by scholars like [Aryabhata](#), [Brahmagupta](#), [Bhaskara II](#), [Varāhamihira](#), and [Madhava](#). The [decimal number system](#) in use today^[3] was first recorded in Indian mathematics.^[4] Indian mathematicians made early contributions to the study of the concept of [zero](#) as a number,^[5] [negative numbers](#),^[6] [arithmetic](#), and [algebra](#).^[7] In addition, [trigonometry](#)^[8] was further advanced in India, and, in particular, the modern definitions of [sine](#) and [cosine](#) were developed there.^[9] These mathematical concepts were transmitted to the Middle East, China, and Europe^[7] and led to further developments that now form the foundations of many areas of mathematics.

Ancient and medieval Indian mathematical works, all composed in [Sanskrit](#), usually consisted of a section of [sutras](#) in which a set of rules or problems were stated with great economy in verse in order to aid memorization by a student. This was followed by a second section consisting of a prose commentary (sometimes multiple commentaries by different scholars) that explained the problem in more detail and provided justification for the solution. In the prose section, the form (and therefore its memorization) was not considered so important as the ideas involved.^{[1][10]} All mathematical works were orally transmitted until approximately 500 BCE; thereafter, they were transmitted both orally and in manuscript form. The oldest extant mathematical document produced on the Indian subcontinent is the birch bark [Bakhshali Manuscript](#), discovered in 1881 in the village of [Bakhshali](#), near [Peshawar](#) (modern day [Pakistan](#)) and is likely from the 7th century CE.^{[11][12]}

A later landmark in Indian mathematics was the development of the [series](#) expansions for [trigonometric functions](#) (sine, cosine, and [arc tangent](#)) by mathematicians of the [Kerala school](#) in the 15th century CE. Their work, completed two centuries before the invention of [calculus](#) in Europe, provided what is now considered the first example of a [power series](#) (apart from geometric series).^[13] However, they did not formulate a systematic theory of [differentiation](#) and [integration](#), nor is there any evidence of their results being transmitted outside [Kerala](#).^{[14][15][16][17]}

Prehistory



Cubical weights standardised in the Indus Valley civilisation

Excavations at [Harappa](#), [Mohenjo-daro](#) and other sites of the [Indus Valley civilisation](#) have uncovered evidence of the use of "practical mathematics". The people of the Indus Valley Civilization manufactured bricks whose dimensions were in the proportion 4:2:1, considered favourable for the stability of a brick structure. They used a standardised system of weights based on the ratios: 1/20, 1/10, 1/5, 1/2, 1, 2, 5, 10, 20, 50, 100, 200, and 500, with the unit weight equaling approximately 28 grams (and approximately equal to the English ounce or Greek uncia). They mass-produced weights in regular [geometrical](#) shapes, which included [hexahedra](#), [barrels](#), [cones](#), and [cylinders](#), thereby demonstrating knowledge of basic [geometry](#).^[18]

The inhabitants of Indus civilisation also tried to standardise measurement of length to a high degree of accuracy. They designed a ruler—the *Mohenjo-daro ruler*—whose unit of length (approximately 1.32 inches or 3.4 centimetres) was divided into ten equal parts. Bricks manufactured in ancient Mohenjo-daro often had dimensions that were integral multiples of this unit of length.^{[19][20]}

Hollow cylindrical objects made of shell and found at [Lothal](#) (2200 BCE) and [Dholavira](#) are demonstrated to have the ability to measure angles in a plane, as well as to determine the position of stars for navigation.^[21]

Vedic period

Samhitas and Brahmanas

The texts of the [Vedic Period](#) provide evidence for the use of [large numbers](#). By the time of the [Yajurvedasamhitā](#)- (1200–900 BCE), numbers as high as 10^{12} were being included in the texts.^[2] For example, the [mantra](#) (sacred recitation) at the end of the *annahoma* ("food-oblation rite") performed

during the [aśvamedha](#), and uttered just before-, during-, and just after sunrise, invokes powers of ten from a hundred to a trillion.^[2]

Hail to *śata* ("hundred," 10^2), hail to *sahasra* ("thousand," 10^3), hail to *ayuta* ("ten thousand," 10^4), hail to *niyuta* ("hundred thousand," 10^5), hail to *prayuta* ("million," 10^6), hail to *arbuda* ("ten million," 10^7), hail to *nyarbuda* ("hundred million," 10^8), hail to *samudra* ("billion," 10^9 , literally "ocean"), hail to *madhya* ("ten billion," 10^{10} , literally "middle"), hail to *anta* ("hundred billion," 10^{11} , lit., "end"), hail to *parārdha* ("one trillion," 10^{12} lit., "beyond parts"), hail to the *uṣas* (dawn), hail to the *vyuṣṭi* (twilight), hail to *udeṣyat* (the one which is going to rise), hail to *udyat* (the one which is rising), hail *udita* (to the one which has just risen), hail to *svarga* (the heaven), hail to *martya* (the world), hail to all.^[2]

Fractions are mentioned, as in the Purusha Sukta (RV 10.90.4):

With three-fourths Puruṣa went up: one-fourth of him again was here.

The [Satapatha Brahmana](#) (c. 7th century BCE) contains rules for ritual geometric constructions that are similar to the Sulba Sūtras.^[22]

Śulba Sūtras

The [Śulba Sūtras](#) (literally, "Aphorisms of the Chords" in [Vedic Sanskrit](#)) (c. 700–400 BCE) list rules for the construction of sacrificial fire altars.^[23] Most mathematical problems considered in the *Śulba Sūtras* spring from "a single theological requirement",^[24] that of constructing fire altars which have different shapes but occupy the same area. The altars were required to be constructed of five layers of burnt brick, with the further condition that each layer consist of 200 bricks and that no two adjacent layers have congruent arrangements of bricks.^[24]

According to Hayashi, the *Śulba Sūtras* contain "the earliest extant verbal expression of the [Pythagorean Theorem](#) in the world, although it had already been known to the [Old Babylonians](#)."

The diagonal rope (*akṣṇayā-rajju*) of an oblong (rectangle) produces both which the flank (*pārśvamāni*) and the horizontal (*tiryāṇmānī*) <ropes> produce separately.^[25]

Since the statement is a *sūtra*, it is necessarily compressed and what the ropes *produce* is not elaborated on, but the context clearly implies the square areas constructed on their lengths, and would have been explained so by the teacher to the student.^[25]

They contain lists of [Pythagorean triples](#),^[26] which are particular cases of [Diophantine equations](#).^[27] They also contain statements (that with hindsight we know to be approximate) about [squaring the circle](#) and "circling the square."^[28]

[Baudhayana](#) (c. 8th century BCE) composed the *Baudhayana Sulba Sutra*, the best-known *Sulba Sutra*, which contains examples of simple Pythagorean triples, such as: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), and (12, 35, 37),^[29] as well as a statement of the Pythagorean theorem for the sides of a square: "The rope which is stretched across the diagonal of a square produces an area double the size of the original square."^{[29][30]} It also contains the general statement of the Pythagorean theorem (for the sides of a rectangle): "The rope stretched along the length of the diagonal of a rectangle makes an area which the vertical and horizontal sides make together."^[29] Baudhayana gives an expression for the [square root of two](#):^[31]

$$\sqrt{2} \approx 1 + \frac{1}{3} + \frac{1}{3 \cdot 4} - \frac{1}{3 \cdot 4 \cdot 34} = 1.4142156 \dots$$

The expression is accurate up to five decimal places, the true value being 1.41421356...^[32] This expression is similar in structure to the expression found on a Mesopotamian tablet^[33] from the Old Babylonian period (1900–1600 [BCE](#)):^[31]

$$\sqrt{2} \approx 1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3} = 1.41421297 \dots$$

which expresses $\sqrt{2}$ in the sexagesimal system, and which is also accurate up to 5 decimal places.

According to mathematician S. G. Dani, the Babylonian cuneiform tablet [Plimpton 322](#) written c. 1850 BCE^[34] "contains fifteen Pythagorean triples with quite large entries, including (13500, 12709, 18541) which is a primitive triple,^[35] indicating, in particular, that there was sophisticated understanding on the topic" in Mesopotamia in 1850 BCE. "Since these tablets predate the *Sulbasutras* period by several centuries, taking into account the contextual appearance of some of the triples, it is reasonable to expect that similar understanding would have been there in India."^[36] Dani goes on to say:

As the main objective of the *Sulvasutras* was to describe the constructions of altars and the geometric principles involved in them, the subject of Pythagorean triples, even if it had been well understood may still not have featured in the *Sulvasutras*. The occurrence of the triples in the *Sulvasutras* is comparable to mathematics that

one may encounter in an introductory book on architecture or another similar applied area, and would not correspond directly to the overall knowledge on the topic at that time. Since, unfortunately, no other contemporaneous sources have been found it may never be possible to settle this issue satisfactorily.^[36]

In all, three *Sulba Sutras* were composed. The remaining two, the *Manava Sulba Sutra* composed by [Manava](#) (fl. 750–650 BCE) and the *Apastamba Sulba Sutra*, composed by [Apastamba](#) (c. 600 BCE), contained results similar to the *Baudhayana Sulba Sutra*.

Vyakarana

The Vedic period saw the work of Sanskrit grammarian [Pāṇini](#) (c. 520–460 BCE). His grammar includes a precursor of the [Backus–Naur form](#) (used in the description [programming languages](#)).^[37]

Pingala (300 BCE – 200 BCE)

Among the scholars of the post-Vedic period who contributed to mathematics, the most notable is [Pingala](#) (*piṅgalā*) (fl. 300–200 BCE), a [music theorist](#) who authored the [Chhandas Shastra](#) (*chandaḥ-sāstra*, also Chhandas Sutra *chandaḥ-sūtra*), a [Sanskrit](#) treatise on [prosody](#). Pingala's work also contains the basic ideas of [Fibonacci numbers](#) (called *maatrameṇu*). Although the *Chandah sutra* has not survived in its entirety, a 10th-century commentary on it by Halāyudha has. Halāyudha, who refers to the Pascal triangle as [Meru-prastāra](#) (literally "the staircase to Mount Meru"), has this to say:

Draw a square. Beginning at half the square, draw two other similar squares below it; below these two, three other squares, and so on. The marking should be started by putting **1** in the first square. Put **1** in each of the two squares of the second line. In the third line put **1** in the two squares at the ends and, in the middle square, the sum of the digits in the two squares lying above it. In the fourth line put **1** in the two squares at the ends. In the middle ones put the sum of the digits in the two squares above each. Proceed in this way. Of these lines, the second gives the combinations with one syllable, the third the combinations with two syllables, ...^[38]

The text also indicates that Pingala was aware of the [combinatorial](#) identity:^[39]

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

Kātyāyana

[Kātyāyana](#) (c. 3rd century BCE) is notable for being the last of the Vedic mathematicians. He wrote the *Katyayana Sulba Sutra*, which presented much [geometry](#), including the general [Pythagorean theorem](#) and a computation of the [square root of 2](#) correct to five decimal places.

Jain mathematics (400 BCE – 200 CE)

Although [Jainism](#) as a religion and philosophy predates its most famous exponent, [Mahavira](#) (6th century BCE), most Jain texts on mathematical topics were composed after the 6th century BCE. [Jain](#) mathematicians are important historically as crucial links between the mathematics of the Vedic period and that of the "classical period."

A significant historical contribution of Jain mathematicians lay in their freeing Indian mathematics from its religious and ritualistic constraints. In particular, their fascination with the enumeration of very large numbers and [infinities](#) led them to classify numbers into three classes: enumerable, innumerable and [infinite](#). Not content with a simple notion of infinity, their texts define five different types of infinity: the infinite in one direction, the infinite in two directions, the infinite in area, the infinite everywhere, and the infinite perpetually. In addition, Jain mathematicians devised notations for simple powers (and exponents) of numbers like squares and cubes, which enabled them to define simple [algebraic equations](#) (*bījagaṇita samīkaraṇa*). Jain mathematicians were apparently also the first to use the word *śhunya* (literally *void* in [Sanskrit](#)) to refer to zero. This word is the ultimate [etymological origin of the English word "zero"](#), as it was [calqued](#) into Arabic as *ṣifr* and then subsequently borrowed into [Medieval Latin](#) as *zephirum*, finally arriving at English after passing through one or more [Romance languages](#) (cf. French *zéro*, Italian *zero*).^[40]

In addition to *Surya Prajnapti*, important Jain works on mathematics included the [Sthānāṅga Sūtra](#) (c. 300 BCE – 200 CE); the *Anuyogadwara Sutra* (c. 200 BCE – 100 CE), which includes the earliest known description of [factorials](#) in Indian mathematics;^[41] and the [Śaṭkhaṇḍāgama](#) (c. 2nd century CE). Important Jain mathematicians included [Bhadrabahu](#) (d. 298 BCE), the author of two astronomical works, the *Bhadrabahavi-Samhita* and a commentary on the *Surya Prajinapti*; Yativrisham Acharya (c. 176 BCE), who authored a mathematical text called [Tiloyapannati](#); and [Umasvati](#) (c. 150 BCE), who, although better known for his influential writings on Jain philosophy and [metaphysics](#), composed a mathematical work called the [Tattvārtha Sūtra](#).

Oral tradition

Mathematicians of ancient and early medieval India were almost all [Sanskrit pandits](#) (*paṇḍita* "learned man"),^[42] who were trained in Sanskrit language and literature, and possessed "a common stock of knowledge in grammar ([vyākaraṇa](#)), [exegesis](#) (*mīmāṃsā*) and logic (*nyāya*)."^[42]

Memorisation of "what is heard" (*śruti* in Sanskrit) through recitation played a major role in the transmission of sacred texts in ancient India. Memorisation and recitation was also used to transmit philosophical and literary works, as well as treatises on ritual and grammar. Modern scholars of ancient India have noted the "truly remarkable achievements of the Indian pandits who have preserved enormously bulky texts orally for millennia."^[43]

Styles of memorisation

Prodigious energy was expended by ancient Indian culture in ensuring that these texts were transmitted from generation to generation with inordinate fidelity.^[44] For example, memorisation of the sacred *Vedas* included up to eleven forms of recitation of the same text. The texts were subsequently "proof-read" by comparing the different recited versions. Forms of recitation included the *jaṭā-pāṭha* (literally "mesh recitation") in which every two adjacent words in the text were first recited in their original order, then repeated in the reverse order, and finally repeated in the original order.^[45] The recitation thus proceeded as:

word1word2, word2word1, word1word2; word2word3, word3word2, word2word3; ...

In another form of recitation, *dhvaja-pāṭha*^[45] (literally "flag recitation") a sequence of N words were recited (and memorised) by pairing the first two and last two words and then proceeding as:

word₁word₂, word_{N-1}word_N; word₂word₃, word_{N-2}word_{N-1}; ...; word_{N-1}word_N, word₁word₂;

The most complex form of recitation, *ghana-pāṭha* (literally "dense recitation"), according to Filliozat,^[45] took the form:

**word1word2, word2word1, word1word2word3, word3word2word1, word1word2word3;
word2word3, word3word2, word2word3word4, word4word3word2, word2word3word4; ...**

That these methods have been effective is testified to by the preservation of the most ancient Indian religious text, the *Rgveda* (c. 1500 BCE), as a single text, without any variant readings.^[45] Similar methods were used for memorising mathematical texts, whose transmission remained exclusively oral until the end of the *Vedic period* (c. 500 BCE).

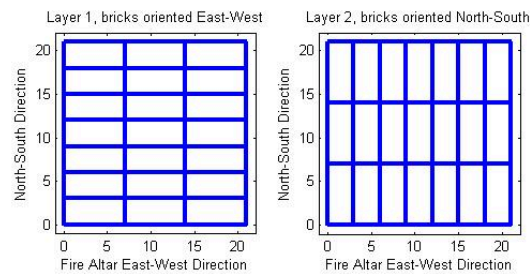
The *Sutra* genre

Mathematical activity in ancient India began as a part of a "methodological reflexion" on the sacred *Vedas*, which took the form of works called *Vedāṅgas*, or, "Ancillaries of the Veda" (7th–4th century BCE).^[46] The need to conserve the sound of sacred text by use of *śikṣā* (phonetics) and *chhandas* (metrics); to conserve its meaning by use of *vyākaraṇa* (grammar) and *nirukta* (etymology); and to

correctly perform the rites at the correct time by the use of *kalpa* (ritual) and *jyotiṣa* (astrology), gave rise to the six disciplines of the *Vedāṅgas*.^[46] Mathematics arose as a part of the last two disciplines, ritual and astronomy (which also included astrology). Since the *Vedāṅgas* immediately preceded the use of writing in ancient India, they formed the last of the exclusively oral literature. They were expressed in a highly compressed mnemonic form, the *sūtra* (literally, "thread"):

The knowers of the *sūtra* know it as having few phonemes, being devoid of ambiguity, containing the essence, facing everything, being without pause and unobjectionable.^[46]

Extreme brevity was achieved through multiple means, which included using *ellipsis* "beyond the tolerance of natural language",^[46] using technical names instead of longer descriptive names, abridging lists by only mentioning the first and last entries, and using markers and variables.^[46] The *sūtras* create the impression that communication through the text was "only a part of the whole instruction. The rest of the instruction must have been transmitted by the so-called *Guru-shishya parampara*, 'uninterrupted succession from teacher (*guru*) to the student (*śisya*)', and it was not open to the general public" and perhaps even kept secret.^[47] The brevity achieved in a *sūtra* is demonstrated in the following example from the Baudhāyana *Śulba Sūtra* (700 BCE).



The design of the domestic fire altar in the *Śulba Sūtra*

The domestic fire-altar in the *Vedic period* was required by ritual to have a square base and be constituted of five layers of bricks with 21 bricks in each layer. One method of constructing the altar was to divide one side of the square into three equal parts using a cord or rope, to next divide the transverse (or perpendicular) side into seven equal parts, and thereby sub-divide the square into 21 congruent rectangles. The bricks were then designed to be of the shape of the constituent rectangle and the layer was created. To form the next layer, the same formula was used, but the bricks were arranged transversely.^[48] The process was then repeated three more times (with alternating

directions) in order to complete the construction. In the Baudhāyana *Śulba Sūtra*, this procedure is described in the following words:

II.64. After dividing the quadri-lateral in seven, one divides the transverse [cord] in three.

II.65. In another layer one places the [bricks] North-pointing.^[48]

According to Filliozat,^[49] the officiant constructing the altar has only a few tools and materials at his disposal: a cord (Sanskrit, *rajju*, f.), two pegs (Sanskrit, *śanku*, m.), and clay to make the bricks (Sanskrit, *iṣṭakā*, f.). Concision is achieved in the *sūtra*, by not explicitly mentioning what the adjective "transverse" qualifies; however, from the feminine form of the (Sanskrit) adjective used, it is easily inferred to qualify "cord." Similarly, in the second stanza, "bricks" are not explicitly mentioned, but inferred again by the feminine plural form of "North-pointing." Finally, the first stanza, never explicitly says that the first layer of bricks are oriented in the east–west direction, but that too is implied by the explicit mention of "North-pointing" in the *second* stanza; for, if the orientation was meant to be the same in the two layers, it would either not be mentioned at all or be only mentioned in the first stanza. All these inferences are made by the officiant as he recalls the formula from his memory.^[48]

The written tradition: prose commentary

With the increasing complexity of mathematics and other exact sciences, both writing and computation were required. Consequently, many mathematical works began to be written down in manuscripts that were then copied and re-copied from generation to generation.

India today is estimated to have about thirty million manuscripts, the largest body of handwritten reading material anywhere in the world. The literate culture of Indian science goes back to at least the fifth century B.C. ... as is shown by the elements of Mesopotamian omen literature and astronomy that entered India at that time and (were) definitely not ... preserved orally.^[50]

The earliest mathematical prose commentary was that on the work, *Āryabhaṭīya* (written 499 CE), a work on astronomy and mathematics. The mathematical portion of the *Āryabhaṭīya* was composed of 33 *sūtras* (in verse form) consisting of mathematical statements or rules, but without any proofs.^[51] However, according to Hayashi,^[52] "this does not necessarily mean that their authors did not prove them. It was probably a matter of style of exposition." From the time of *Bhaskara I* (600 CE

onwards), prose commentaries increasingly began to include some derivations (*upapatti*). Bhaskara I's commentary on the *Āryabhaṭīya*, had the following structure:^[51]

- **Rule** ('sūtra') in verse by *Āryabhaṭa*
- **Commentary** by Bhāskara I, consisting of:
 - **Elucidation** of rule (derivations were still rare then, but became more common later)
 - **Example** (*uddeśaka*) usually in verse.
 - **Setting** (*nyāsa/sthāpanā*) of the numerical data.
 - **Working** (*karana*) of the solution.
 - **Verification** (*pratyayakaraṇa*, literally "to make conviction") of the answer. These became rare by the 13th century, derivations or proofs being favoured by then.^[51]

Typically, for any mathematical topic, students in ancient India first memorised the *sūtras*, which, as explained earlier, were "deliberately inadequate"^[50] in explanatory details (in order to pithily convey the bare-bone mathematical rules). The students then worked through the topics of the prose commentary by writing (and drawing diagrams) on chalk- and dust-boards (*i.e.* boards covered with dust). The latter activity, a staple of mathematical work, was to later prompt mathematician-astronomer, *Brahmagupta* (fl. 7th century CE), to characterise astronomical computations as "dust work" (Sanskrit: *dhulikarman*).^[53]

Numerals and the decimal number system

It is well known that the decimal place-value system *in use today* was first recorded in India, then transmitted to the Islamic world, and eventually to Europe.^[54] The Syrian bishop *Severus Sebokht* wrote in the mid-7th century CE about the "nine signs" of the Indians for expressing numbers.^[54] However, how, when, and where the first decimal place value system was invented is not so clear.^[55]

The earliest extant *script* used in India was the *Kharoṣṭhī* script used in the *Gandhara* culture of the north-west. It is thought to be of *Aramaic* origin and it was in use from the 4th century BCE to the 4th century CE. Almost contemporaneously, another script, the *Brāhmī script*, appeared on much of the sub-continent, and would later become the foundation of many scripts of South Asia and South-east Asia. Both scripts had numeral symbols and numeral systems, which were initially *not* based on a place-value system.^[56]

The earliest surviving evidence of decimal place value numerals in India and southeast Asia is from the middle of the first millennium CE.^[57] A copper plate from Gujarat, India mentions the date 595 CE, written in a decimal place value notation, although there is some doubt as to the

authenticity of the plate.^[57] Decimal numerals recording the years 683 CE have also been found in stone inscriptions in Indonesia and Cambodia, where Indian cultural influence was substantial.^[57]

There are older textual sources, although the extant manuscript copies of these texts are from much later dates.^[58] Probably the earliest such source is the work of the Buddhist philosopher Vasumitra dated likely to the 1st century CE.^[58] Discussing the counting pits of merchants, Vasumitra remarks, "When [the same] clay counting-piece is in the place of units, it is denoted as one, when in hundreds, one hundred."^[58] Although such references seem to imply that his readers had knowledge of a decimal place value representation, the "brevity of their allusions and the ambiguity of their dates, however, do not solidly establish the chronology of the development of this concept."^[58]

A third decimal representation was employed in a verse composition technique, later labelled *Bhuta-sankhya* (literally, "object numbers") used by early Sanskrit authors of technical books.^[59] Since many early technical works were composed in verse, numbers were often represented by objects in the natural or religious world that corresponded to them; this allowed a many-to-one correspondence for each number and made verse composition easier.^[59] According to Plofker,^[60] the number 4, for example, could be represented by the word "*Veda*" (since there were four of these religious texts), the number 32 by the word "teeth" (since a full set consists of 32), and the number 1 by "moon" (since there is only one moon).^[59] So, *Veda*/teeth/moon would correspond to the decimal numeral 1324, as the convention for numbers was to enumerate their digits from right to left.^[59] The earliest reference employing object numbers is a c. 269 CE Sanskrit text, *Yavanajātaka* (literally "Greek horoscopy") of Sphujidhvaja, a versification of an earlier (c. 150 CE) Indian prose adaptation of a lost work of Hellenistic astrology.^[61] Such use seems to make the case that by the mid-3rd century CE, the decimal place value system was familiar, at least to readers of astronomical and astrological texts in India.^[59]

It has been hypothesized that the Indian decimal place value system was based on the symbols used on Chinese counting boards from as early as the middle of the first millennium BCE.^[62] According to Plofker,^[60]

These counting boards, like the Indian counting pits, ..., had a decimal place value structure ... Indians may well have learned of these decimal place value "rod numerals" from Chinese Buddhist pilgrims or other travelers, or they may have developed the concept independently from their earlier non-place-value system; no documentary evidence survives to confirm either conclusion."^[62]

Bakhshali Manuscript

The oldest extant mathematical manuscript in India is the *Bakhshali Manuscript*, a birch bark manuscript written in "Buddhist hybrid Sanskrit"^[12] in the *Śāradā* script, which was used in the northwestern region of the Indian subcontinent between the 8th and 12th centuries CE.^[63] The manuscript was discovered in 1881 by a farmer while digging in a stone enclosure in the village of Bakhshali, near [Peshawar](#) (then in [British India](#) and now in [Pakistan](#)). Of unknown authorship and now preserved in the [Bodleian Library](#) in the [University of Oxford](#), the manuscript has been dated recently as 224–383 CE.^[64]

The surviving manuscript has seventy leaves, some of which are in fragments. Its mathematical content consists of rules and examples, written in verse, together with prose commentaries, which include solutions to the examples.^[63] The topics treated include arithmetic (fractions, square roots, profit and loss, simple interest, the [rule of three](#), and *regula falsi*) and algebra (simultaneous linear equations and [quadratic equations](#)), and arithmetic progressions. In addition, there is a handful of geometric problems (including problems about volumes of irregular solids). The Bakhshali manuscript also "employs a decimal place value system with a dot for zero."^[63] Many of its problems are of a category known as 'equalisation problems' that lead to systems of linear equations. One example from Fragment III-5-3v is the following:

One merchant has seven *asava* horses, a second has nine *haya* horses, and a third has ten camels. They are equally well off in the value of their animals if each gives two animals, one to each of the others. Find the price of each animal and the total value for the animals possessed by each merchant.^[65]

The prose commentary accompanying the example solves the problem by converting it to three (under-determined) equations in four unknowns and assuming that the prices are all integers.^[65]

In 2017, three samples from the manuscript were shown by [radiocarbon dating](#) to come from three different centuries: from 224 to 383 CE, 680–779 CE, and 885–993 CE. It is not known how fragments from different centuries came to be packaged together.^{[66][67][68]}

Classical period (400–1300)

This period is often known as the golden age of Indian Mathematics. This period saw mathematicians such as [Aryabhata](#), [Varahamihira](#), [Brahmagupta](#), [Bhaskara I](#), [Mahavira](#), [Bhaskara II](#), [Madhava of Sangamagrama](#) and [Nilakantha Somayaji](#) give broader and clearer shape to many

branches of mathematics. Their contributions would spread to Asia, the Middle East, and eventually to Europe. Unlike Vedic mathematics, their works included both astronomical and mathematical contributions. In fact, mathematics of that period was included in the 'astral science' (*jyotiḥśāstra*) and consisted of three sub-disciplines: mathematical sciences (*gaṇita* or *tantra*), horoscope astrology (*horā* or *jātaka*) and divination (*saṃhitā*).^[53] This tripartite division is seen in Varāhamihira's 6th century compilation—*Pancasiddhantika*^[69] (literally *panca*, "five", *siddhānta*, "conclusion of deliberation", dated 575 CE)—of five earlier works, *Surya Siddhanta*, *Romaka Siddhanta*, *Paulisa Siddhanta*, *Vasishtha Siddhanta* and *Paitamaha Siddhanta*, which were adaptations of still earlier works of Mesopotamian, Greek, Egyptian, Roman and Indian astronomy. As explained earlier, the main texts were composed in Sanskrit verse, and were followed by prose commentaries.^[53]

Fourth to sixth centuries

Surya Siddhanta

Though its authorship is unknown, the *Surya Siddhanta* (c. 400) contains the roots of modern *trigonometry*. Because it contains many words of foreign origin, some authors consider that it was written under the influence of *Mesopotamia* and Greece.^[70]

This ancient text uses the following as trigonometric functions for the first time:

- Sine (*Jya*).
- Cosine (*Kojya*).
- Inverse sine (*Otkram jya*).

Later Indian mathematicians such as Aryabhata made references to this text, while later *Arabic* and *Latin* translations were very influential in Europe and the Middle East.

Chhedi calendar

This Chhedi calendar (594) contains an early use of the modern *place-value Hindu–Arabic numeral system* now used universally.

Aryabhata I

Aryabhata (476–550) wrote the *Aryabhatiya*. He described the important fundamental principles of mathematics in 332 *shlokas*. The treatise contained:

- Quadratic equations
- Trigonometry

- The value of π , correct to 4 decimal places.

Aryabhata also wrote the *Arya Siddhanta*, which is now lost. Aryabhata's contributions include:

Trigonometry:

(See also : [Aryabhata's sine table](#))

- Introduced the [trigonometric functions](#).
- Defined the sine (*jya*) as the modern relationship between half an angle and half a chord.
- Defined the cosine (*kojya*).
- Defined the [versine](#) (*utkrama-jya*).
- Defined the inverse sine (*otkram jya*).
- Gave methods of calculating their approximate numerical values.
- Contains the earliest tables of sine, cosine and versine values, in 3.75° intervals from 0° to 90°, to 4 decimal places of accuracy.
- Contains the trigonometric formula $\sin(n + 1)x - \sin nx = \sin nx - \sin(n - 1)x - (1/225)\sin nx$.
- [Spherical trigonometry](#).

Arithmetic:

- [Simple continued fractions](#).

Algebra:

- Solutions of simultaneous quadratic equations.
- Whole number solutions of [linear equations](#) by a method equivalent to the modern method.
- General solution of the indeterminate linear equation .

Mathematical astronomy:

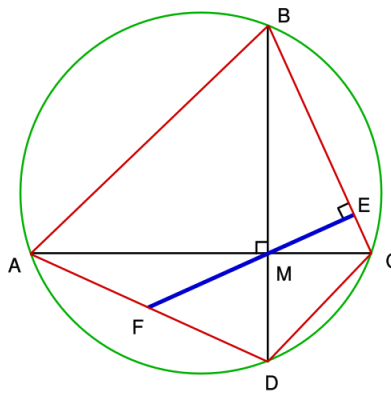
- Accurate calculations for astronomical constants, such as the:
 - [Solar eclipse](#).
 - [Lunar eclipse](#).
 - The formula for the sum of the [cubes](#), which was an important step in the development of integral calculus.^[71]

Varahamihira

Varahamihira (505–587) produced the *Pancha Siddhanta* (*The Five Astronomical Canons*). He made important contributions to trigonometry, including sine and cosine tables to 4 decimal places of accuracy and the following formulas relating **sine** and **cosine** functions:

- $\sin^2(x) + \cos^2(x) = 1$
- $\sin(x) = \cos\left(\frac{\pi}{2} - x\right)$
- $\frac{1 - \cos(2x)}{2} = \sin^2(x)$

Seventh and eighth centuries



Brahmagupta's theorem states that $AF = FD$.

In the 7th century, two separate fields, **arithmetic** (which included **measurement**) and **algebra**, began to emerge in Indian mathematics. The two fields would later be called *pāṭī-gaṇita* (literally "mathematics of algorithms") and *bījā-gaṇita* (lit. "mathematics of seeds", with "seeds"—like the seeds of plants—representing unknowns with the potential to generate, in this case, the solutions of equations).^[72] **Brahmagupta**, in his astronomical work *Brāhma Sphuṭa Siddhānta* (628 CE), included two chapters (12 and 18) devoted to these fields. Chapter 12, containing 66 Sanskrit verses, was divided into two sections: "basic operations" (including cube roots, fractions, ratio and proportion, and barter) and "practical mathematics" (including mixture, mathematical series, plane figures, stacking bricks, sawing of timber, and piling of grain).^[73] In the latter section, he stated his famous theorem on the diagonals of a **cyclic quadrilateral**:^[73]

Brahmagupta's theorem: If a cyclic quadrilateral has diagonals that are **perpendicular** to each other, then the perpendicular line drawn from the point of intersection of the diagonals to any side of the quadrilateral always bisects the opposite side.

Chapter 12 also included a formula for the area of a cyclic quadrilateral (a generalisation of [Heron's formula](#)), as well as a complete description of [rational triangles](#) (i.e. triangles with rational sides and rational areas).

Brahmagupta's formula: The area, A , of a cyclic quadrilateral with sides of lengths a, b, c, d , respectively, is given by

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where s , the [semiperimeter](#), given by $s = \frac{a+b+c+d}{2}$.

Brahmagupta's Theorem on rational triangles: A triangle with rational sides a, b, c and rational area is of the form:

$$a = \frac{u^2}{v} + v, \quad b = \frac{u^2}{w} + w, \quad c = \frac{u^2}{v} + \frac{u^2}{w} - (v + w)$$

for some rational numbers u, v , and w .^[74]

Chapter 18 contained 103 Sanskrit verses which began with rules for arithmetical operations involving zero and negative numbers^[73] and is considered the first systematic treatment of the subject. The rules (which included $a + 0 = a$ and $a \times 0 = 0$) were all correct, with one exception: $\frac{0}{0} = 0$.^[73] Later in the chapter, he gave the first explicit (although still not completely general) solution of the [quadratic equation](#):

$$ax^2 + bx = c$$

To the absolute number multiplied by four times the [coefficient of the] square, add the square of the [coefficient of the] middle term; the square root of the same, less the [coefficient of the] middle term, being divided by twice the [coefficient of the] square is the value.^[75]

This is equivalent to:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

Also in chapter 18, Brahmagupta was able to make progress in finding (integral) solutions of [Pell's equation](#),^[76]

$$x^2 - Ny^2 = 1,$$

where N is a nonsquare integer. He did this by discovering the following identity:^[76]

Brahmagupta's Identity: $(x^2 - Ny^2)(x'^2 - Ny'^2) = (xx' + Ny y')^2 - N(xy' + x'y)^2$

which was a generalisation of an earlier identity of [Diophantus](#):^[76] Brahmagupta used his identity to prove the following lemma:^[76]

Lemma (Brahmagupta): If $x = x_1, y = y_1$ is a solution of $x^2 - Ny^2 = k_1$, and, $x = x_2, y = y_2$ is a solution of $x^2 - Ny^2 = k_2$, then:

$$x = x_1 x_2 + N y_1 y_2, \quad y = x_1 y_2 + x_2 y_1 \quad \text{is a solution of } x^2 - Ny^2 = k_1 k_2$$

He then used this lemma to both generate infinitely many (integral) solutions of Pell's equation, given one solution, and state the following theorem:

Theorem (Brahmagupta): If the equation $x^2 - Ny^2 = k$ has an integer solution for any one of $k = \pm 4, \pm 2, -1$ then Pell's equation:

$$x^2 - Ny^2 = 1$$

also has an integer solution.^[77]

Brahmagupta did not actually prove the theorem, but rather worked out examples using his method. The first example he presented was:^[76]

Example (Brahmagupta): Find integers x, y such that:

$$x^2 - 92y^2 = 1$$

In his commentary, Brahmagupta added, "a person solving this problem within a year is a mathematician."^[76] The solution he provided was:

$$x = 1151, \quad y = 120$$

Bhaskara I

[Bhaskara I](#) (c. 600–680) expanded the work of Aryabhata in his books titled *Mahabhaskariya*, *Aryabhatiya-bhashya* and *Laghu-bhaskariya*. He produced:

- Solutions of indeterminate equations.
- A rational approximation of the [sine function](#).
- A formula for calculating the sine of an acute angle without the use of a table, correct to two decimal places.

Ninth to twelfth centuries

Virasena

[Virasena](#) (8th century) was a Jain mathematician in the court of [Rashtrakuta](#) King [Amoghavarsha](#) of [Manyakheta](#), Karnataka. He wrote the *Dhavalā*, a commentary on Jain mathematics, which:

- Deals with the concept of *ardhaccheda*, the number of times a number could be halved, and lists various rules involving this operation. This coincides with the [binary logarithm](#) when applied to [powers of two](#),^{[78][79]} but differs on other numbers, more closely resembling the [2-adic order](#).

Virasena also gave:

- The derivation of the [volume](#) of a [frustum](#) by a sort of infinite procedure.

It is thought that much of the mathematical material in the *Dhavalā* can attributed to previous writers, especially Kundakunda, Shamakunda, Tumbulura, Samantabhadra and Bappadeva and date who wrote between 200 and 600 CE.^[79]

Mahavira

[Mahavira Acharya](#) (c. 800–870) from [Karnataka](#), the last of the notable Jain mathematicians, lived in the 9th century and was patronised by the Rashtrakuta king Amoghavarsha. He wrote a book titled *Ganit Saar Sangraha* on numerical mathematics, and also wrote treatises about a wide range of mathematical topics. These include the mathematics of:

- [Zero](#)
- [Squares](#)
- [Cubes](#)
- [square roots](#), [cube roots](#), and the [series](#) extending beyond these
- Plane geometry
- [Solid geometry](#)
- Problems relating to the casting of [shadows](#)
- Formulae derived to calculate the area of an [ellipse](#) and [quadrilateral](#) inside a [circle](#).

Mahavira also:

- Asserted that the square root of a [negative number](#) did not exist
- Gave the sum of a series whose terms are [squares](#) of an [arithmetical progression](#), and gave empirical rules for area and [perimeter](#) of an ellipse.

- Solved cubic equations.
- Solved quartic equations.
- Solved some [quintic equations](#) and higher-order [polynomials](#).
- Gave the general solutions of the higher order polynomial equations:
 - $ax^n = q$
 - $a \frac{x^n - 1}{x - 1} = p$
- Solved indeterminate quadratic equations.
- Solved indeterminate cubic equations.
- Solved indeterminate higher order equations.

Shridhara

[Shridhara](#) (c. 870–930), who lived in [Bengal](#), wrote the books titled *Nav Shatika*, *Tri Shatika* and *Pati Ganita*. He gave:

- A good rule for finding the volume of a [sphere](#).
- The formula for solving [quadratic equations](#).

The *Pati Ganita* is a work on arithmetic and [measurement](#). It deals with various operations, including:

- Elementary operations
- Extracting square and cube roots.
- Fractions.
- Eight rules given for operations involving zero.
- Methods of [summation](#) of different arithmetic and geometric series, which were to become standard references in later works.

Manjula

Aryabhata's equations were elaborated in the 10th century by Manjula (also *Munjala*), who realised that the expression^[80]

$$\sin w' - \sin w$$

could be approximately expressed as

$$(w' - w) \cos w$$

This was elaborated by his later successor Bhaskara ii thereby finding the derivative of sine.^[80]

Aryabhata II

[Aryabhata II](#) (c. 920–1000) wrote a commentary on Shridhara, and an astronomical treatise *Maha-Siddhanta*. The Maha-Siddhanta has 18 chapters, and discusses:

- Numerical mathematics (*Ank Ganit*).
- Algebra.
- Solutions of indeterminate equations (*kuttaka*).

Shripati

[Shripati Mishra](#) (1019–1066) wrote the books *Siddhanta Shekhara*, a major work on astronomy in 19 chapters, and *Ganit Tilaka*, an incomplete [arithmetical](#) treatise in 125 verses based on a work by Shridhara. He worked mainly on:

- [Permutations and combinations](#).
- General solution of the simultaneous indeterminate linear equation.

He was also the author of *Dhikotidakarana*, a work of twenty verses on:

- [Solar eclipse](#).
- [Lunar eclipse](#).

The *Dhruvamanasa* is a work of 105 verses on:

- Calculating planetary [longitudes](#)
- [eclipses](#).
- planetary [transits](#).

Nemichandra Siddhanta Chakravati

Nemichandra Siddhanta Chakravati (c. 1100) authored a mathematical treatise titled *Gome-mat Saar*.

Bhaskara II

[Bhāskara II](#) (1114–1185) was a mathematician-astronomer who wrote a number of important treatises, namely the *Siddhanta Shiromani*, *Lilavati*, *Bijaganita*, *Gola Addhaya*, *Griha Ganitam* and *Karan Kautoohal*. A number of his contributions were later transmitted to the Middle East and Europe. His contributions include:

Arithmetic:

- Interest computation
- Arithmetical and geometrical progressions
- Plane geometry
- Solid geometry
- The shadow of the [gnomon](#)
- Solutions of [combinations](#)
- Gave a proof for division by zero being [infinity](#).

Algebra:

- The recognition of a positive number having two square roots.
- [Surds](#).
- Operations with products of several unknowns.
- The solutions of:
 - Quadratic equations.
 - Cubic equations.
 - Quartic equations.
 - Equations with more than one unknown.
 - Quadratic equations with more than one unknown.
 - The general form of [Pell's equation](#) using the [chakravala method](#).
 - The general indeterminate quadratic equation using the *chakravala* method.
 - Indeterminate cubic equations.
 - Indeterminate quartic equations.
 - Indeterminate higher-order polynomial equations.

Geometry:

- Gave a proof of the [Pythagorean theorem](#).

Calculus:

- Preliminary concept of differentiation
- Discovered the [differential coefficient](#).

- Stated early form of [Rolle's theorem](#), a special case of the [mean value theorem](#) (one of the most important theorems of calculus and analysis).
- Derived the differential of the sine function although did not perceive the notion of derivative.
- Computed π , correct to five decimal places.
- Calculated the length of the Earth's revolution around the Sun to 9 decimal places.^[81]

Trigonometry:

- Developments of [spherical trigonometry](#)
- The trigonometric formulas:
 - $\sin(a + b) = \sin(a) \cos(b) + \sin(b) \cos(a)$
 - $\sin(a - b) = \sin(a) \cos(b) - \sin(b) \cos(a)$

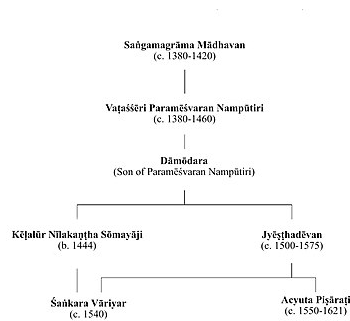
Medieval and early modern mathematics (1300–1800)

Navya-Nyaya

The Navya-Nyāya or Neo-Logical darśana (school) of Indian philosophy was founded in the 13th century by the philosopher [Gangesha Upadhyaya](#) of [Mithila](#).^[82] It was a development of the classical Nyāya darśana. Other influences on Navya-Nyāya were the work of earlier philosophers [Vācaspati Miśra](#) (900–980 CE) and [Udayana](#) (late 10th century).

Gangeśa's book [Tattvacintāmaṇi](#) ("Thought-Jewel of Reality") was written partly in response to Śrīharśa's *Khandanakhandakhādyā*, a defence of [Advaita Vedānta](#), which had offered a set of thorough criticisms of Nyāya theories of thought and language.^[83] Navya-Nyāya developed a sophisticated language and conceptual scheme that allowed it to raise, analyze, and solve problems in logic and epistemology. It involves naming each object to be analyzed, identifying a distinguishing characteristic for the named object, and verifying the appropriateness of the defining characteristic using *pramanas*.^[84]

Kerala School



Chain of teachers of [Kerala school of astronomy and mathematics](#)



Pages from the [Yuktibhāṣā](#) c.1530

The [Kerala school of astronomy and mathematics](#) was founded by [Madhava of Sangamagrama](#) in Kerala, [South India](#) and included among its members: [Parameshvara](#), [Neelakanta Somayaji](#), [Jyesthadeva](#), [Achyuta Pisharati](#), [Melpathur Narayana Bhattathiri](#) and Achyuta Panikkar. It flourished between the 14th and 16th centuries and the original discoveries of the school seems to have ended with Narayana Bhattathiri (1559–1632). In attempting to solve astronomical problems, the Kerala school astronomers *independently* created a number of important mathematics concepts. The most important results, series expansion for [trigonometric functions](#), were given in [Sanskrit](#) verse in a book by Neelakanta called *Tantrasangraha* and a commentary on this work called *Tantrasangraha-vakhya* of unknown authorship. The theorems were stated without proof, but proofs for the series for *sine*, *cosine*, and inverse *tangent* were provided a century later in the work [Yuktibhāṣā](#) (c.1500–c.1610), written in [Malayalam](#), by [Jyesthadeva](#).^[85]

Their discovery of these three important series expansions of [calculus](#)—several centuries before calculus was developed in Europe by [Isaac Newton](#) and [Gottfried Leibniz](#)—was an achievement. However, the Kerala School did not invent *calculus*,^[86] because, while they were able to develop [Taylor series](#) expansions for the important [trigonometric functions](#), they developed neither a theory of [differentiation](#) or [integration](#), nor the [fundamental theorem of calculus](#).^[71] The results obtained by the Kerala school include:

- The (infinite) [geometric series](#): $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$ for $|x| < 1$ ^[87]

- A semi-rigorous proof (see "induction" remark below) of the result:

$$1^p + 2^p + \dots + n^p \approx \frac{n^{p+1}}{p+1} \text{ for large } n. [85]$$

- Intuitive use of [mathematical induction](#), however, the *inductive hypothesis* was not formulated or employed in proofs. [85]
- Applications of ideas from (what was to become) differential and integral calculus to obtain [\(Taylor–Maclaurin\) infinite series](#) for $\sin x$, $\cos x$, and $\arctan x$. [86] The *Tantrasangraha-vakhya* gives the series in verse, which when translated to mathematical notation, can be written as: [85]

$$r \arctan\left(\frac{y}{x}\right) = \frac{1}{1} \cdot \frac{ry}{x} - \frac{1}{3} \cdot \frac{ry^3}{x^3} + \frac{1}{5} \cdot \frac{ry^5}{x^5} - \dots, \text{ where } y/x \leq 1.$$

$$r \sin x = x - x \frac{x^2}{(2^2 + 2)r^2} + x \frac{x^2}{(2^2 + 2)r^2} \cdot \frac{x^2}{(4^2 + 4)r^2} - \dots$$

$$r - \cos x = r \frac{x^2}{(2^2 - 2)r^2} - r \frac{x^2}{(2^2 - 2)r^2} \frac{x^2}{(4^2 - 4)r^2} + \dots,$$

where, for $r = 1$, the series reduces to the standard power series for these trigonometric functions, for example:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

and

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

- Use of rectification (computation of length) of the arc of a circle to give a proof of these results. (The later method of Leibniz, using quadrature, *i.e.* computation of *area under* the arc of the circle, was *not* used.) [85]
- Use of the series expansion of **$\arctan x$** to obtain the [Leibniz formula for \$\pi\$](#) : [85]

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

- A rational approximation of *error* for the finite sum of their series of interest. For example, the error, $f_i(n+1)$, (for n odd, and $i = 1, 2, 3$) for the series:

$$\frac{\pi}{4} \approx 1 - \frac{1}{3} + \frac{1}{5} - \dots + (-1)^{(n-1)/2} \frac{1}{n} + (-1)^{(n+1)/2} f_i(n+1)$$

$$\text{where } f_1(n) = \frac{1}{2n}, f_2(n) = \frac{n/2}{n^2 + 1}, f_3(n) = \frac{(n/2)^2 + 1}{(n^2 + 5)n/2}.$$

- Manipulation of error term to derive a faster converging series for π : [85]

$$\frac{\pi}{4} = \frac{3}{4} + \frac{1}{3^3 - 3} - \frac{1}{5^3 - 5} + \frac{1}{7^3 - 7} - \dots$$

- Using the improved series to derive a rational expression,^[85] 104348/33215 for π correct up to nine decimal places, i.e. 3.141592653.
- Use of an intuitive notion of limit to compute these results.^[85]
- A semi-rigorous (see remark on limits above) method of differentiation of some trigonometric functions.^[71] However, they did not formulate the notion of a *function*, or have knowledge of the exponential or logarithmic functions.

The works of the Kerala school were first written up for the Western world by Englishman [C.M. Whish](#) in 1835. According to Whish, the Kerala mathematicians had "*laid the foundation for a complete system of fluxions*" and these works abounded "*with fluxional forms and series to be found in no work of foreign countries.*"^[88]

However, Whish's results were almost completely neglected, until over a century later, when the discoveries of the Kerala school were investigated again by C. Rajagopal and his associates. Their work includes commentaries on the proofs of the arctan series in *Yuktibhāṣā* given in two papers,^{[89][90]} a commentary on the *Yuktibhāṣā*'s proof of the sine and cosine series^[91] and two papers that provide the Sanskrit verses of the *Tantrasangrahavakhya* for the series for arctan, sin, and cosine (with English translation and commentary).^{[92][93]}

Parameshvara (c. 1370–1460) wrote commentaries on the works of [Bhaskara I](#), [Aryabhata](#) and Bhaskara II. His *Lilavati Bhasya*, a commentary on Bhaskara II's *Lilavati*, contains one of his important discoveries: a version of the [mean value theorem](#). [Nilakantha Somayaji](#) (1444–1544) composed the *Tantra Samgraha* (which 'spawned' a later anonymous commentary *Tantrasangrahavyakhya* and a further commentary by the name *Yuktidipaika*, written in 1501). He elaborated and extended the contributions of Madhava.

[Citrabhanu](#) (c. 1530) was a 16th-century mathematician from Kerala who gave integer solutions to 21 types of systems of two [simultaneous](#) algebraic equations in two unknowns. These types are all the possible pairs of equations of the following seven forms:

$$x + y = a, \quad x - y = b, \quad xy = c, \quad x^2 + y^2 = d,$$

$$x^2 - y^2 = e, \quad x^3 + y^3 = f, \quad x^3 - y^3 = g$$

For each case, Citrabhanu gave an explanation and justification of his rule as well as an example. Some of his explanations are algebraic, while others are geometric. [Jyesthadeva](#) (c. 1500–1575) was another member of the Kerala School. His key work was the *Yukti-bhāṣā* (written in Malayalam,

a regional language of Kerala). Jyesthadeva presented proofs of most mathematical theorems and infinite series earlier discovered by Madhava and other Kerala School mathematicians.

Others

[Narayana Pandit](#) was a 14th century mathematician who composed two important mathematical works, an arithmetical treatise, *Ganita Kaumudi*, and an algebraic treatise, *Bijganita Vatamsa*. *Ganita Kaumudi* is one of the most revolutionary works in the field of combinatorics with developing a method for [systematic generation of all permutations](#) of a given sequence. In his *Ganita Kaumudi* Narayana proposed the following problem on a herd of cows and calves:

A cow produces one calf every year. Beginning in its fourth year, each calf produces one calf at the beginning of each year. How many cows and calves are there altogether after 20 years?

Translated into the modern mathematical language of [recurrence sequences](#):

$$N_n = N_{n-1} + N_{n-3} \text{ for } n > 2,$$

with initial values

$$N_0 = N_1 = N_2 = 1.$$

The first few terms are 1, 1, 1, 2, 3, 4, 6, 9, 13, 19, 28, 41, 60, 88,... (sequence [A000930](#) in the [OEIS](#)).

The limit ratio between consecutive terms is the [supergolden ratio](#). . Narayana is also thought to be the author of an elaborate commentary of [Bhaskara II's Lilavati](#), titled *Ganita Kaumudia*(or *Karma-Paddhati*).^[94]

Charges of Eurocentrism

It has been suggested that Indian contributions to mathematics have not been given due acknowledgement in modern history and that many discoveries and inventions by [Indian mathematicians](#) are presently culturally attributed to their [Western](#) counterparts, as a result of [Eurocentrism](#). According to G. G. Joseph's take on "[Ethnomathematics](#)":

[Their work] takes on board some of the objections raised about the classical Eurocentric trajectory. The awareness [of Indian and Arabic mathematics] is all too likely to be tempered with dismissive rejections of their importance compared to Greek mathematics. The contributions from

other civilisations – most notably China and India, are perceived either as borrowers from Greek sources or having made only minor contributions to mainstream mathematical development. An openness to more recent research findings, especially in the case of Indian and Chinese mathematics, is sadly missing"^[95]

Historian of mathematics [Florian Cajori](#) wrote that he and others "suspect that [Diophantus](#) got his first glimpse of algebraic knowledge from India".^[96] He also wrote that "it is certain that portions of Hindu mathematics are of Greek origin".^[97]

More recently, as discussed in the above section, the infinite series of [calculus](#) for trigonometric functions (rediscovered by Gregory, Taylor, and Maclaurin in the late 17th century) were described in India, by mathematicians of the [Kerala school](#), some two centuries earlier. Some scholars have recently suggested that knowledge of these results might have been transmitted to Europe through the trade route from [Kerala](#) by traders and [Jesuit](#) missionaries.^[98] Kerala was in continuous contact with China and [Arabia](#), and, from around 1500, with Europe. The fact that the communication routes existed and the chronology is suitable certainly make such transmission a possibility. However, no evidence of transmission has been found.^[98] According to [David Bressoud](#), "there is no evidence that the Indian work of series was known beyond India, or even outside of Kerala, until the nineteenth century".^{[86][99]}

Both Arab and Indian scholars made discoveries before the 17th century that are now considered a part of calculus.^[71] However, they did not (as [Newton](#) and [Leibniz](#) did) "combine many differing ideas under the two unifying themes of the [derivative](#) and the [integral](#), show the connection between the two, and turn calculus into the great problem-solving tool we have today".^[71] The intellectual careers of both Newton and Leibniz are well-documented and there is no indication of their work not being their own;^[71] however, it is not known with certainty whether the immediate *predecessors* of Newton and Leibniz, "including, in particular, Fermat and Roberval, [may have] learned of some of the ideas of the Islamic and Indian mathematicians through sources we are not now aware."^[71] This is an area of current research, especially in the manuscript collections of Spain and [Maghreb](#), and is being pursued, among other places, at the [CNRS](#).^[71]

See also

- [Shulba Sutras](#)
- [Kerala school of astronomy and mathematics](#)

- [Surya Siddhanta](#)
- [Brahmagupta](#)
- [Srinivasa Ramanujan](#)
- [Bakhshali manuscript](#)
- [List of Indian mathematicians](#)
- [Indian science and technology](#)
- [Indian logic](#)
- [Indian astronomy](#)
- [History of mathematics](#)
- [List of numbers in Hindu scriptures](#)

Notes

1. ([Kim Plofker 2007](#), p. 1)
2. ([Hayashi 2005](#), pp. 360–361)
3. ([Ifrah 2000](#), p. 346): "The measure of the genius of Indian civilisation, to which we owe our modern (number) system, is all the greater in that it was the only one in all history to have achieved this triumph. Some cultures succeeded, earlier than the Indian, in discovering one or at best two of the characteristics of this intellectual feat. But none of them managed to bring together into a complete and coherent system the necessary and sufficient conditions for a number-system with the same potential as our own."
4. ([Plofker 2009](#), pp. 44–47)
5. ([Bourbaki 1998](#), p. 46): "...our decimal system, which (by the agency of the Arabs) is derived from Hindu mathematics, where its use is attested already from the first centuries of our era. It must be noted moreover that the conception of zero as a number and not as a simple symbol of separation) and its introduction into calculations, also count amongst the original contribution of the Hindus."

6. (Bourbaki 1998, p. 49): Modern arithmetic was known during medieval times as "Modus Indorum" or method of the Indians. Leonardo of Pisa wrote that compared to method of the Indians all other methods is a mistake. This method of the Indians is none other than our very simple arithmetic of addition, subtraction, multiplication and division. Rules for these four simple procedures was first written down by Brahmagupta during the 7th century AD. "On this point, the Hindus are already conscious of the interpretation that negative numbers must have in certain cases (a debt in a commercial problem, for instance). In the following centuries, as there is a diffusion into the West (by intermediary of the Arabs) of the methods and results of Greek and Hindu mathematics, one becomes more used to the handling of these numbers, and one begins to have other "representation" for them which are geometric or dynamic."
7. "algebra" 2007. *Britannica Concise Encyclopedia* (<https://www.britannica.com/ebc/article-231064>) Archived (<https://web.archive.org/web/20070929134632/http://www.britannica.com/ebc/article-231064>) 29 September 2007 at the Wayback Machine. Encyclopædia Britannica Online. 16 May 2007. Quote: "A full-fledged decimal, positional system certainly existed in India by the 9th century (AD), yet many of its central ideas had been transmitted well before that time to China and the Islamic world. Indian arithmetic, moreover, developed consistent and correct rules for operating with positive and negative numbers and for treating zero like any other number, even in problematic contexts such as division. Several hundred years passed before European mathematicians fully integrated such ideas into the developing discipline of algebra."
8. (Pingree 2003, p. 45) Quote: "Geometry, and its branch trigonometry, was the mathematics Indian astronomers used most frequently. Greek mathematicians used the full chord and never imagined the half chord that we use today. Half chord was first used by Aryabhata which made trigonometry much more simple. In fact, the Indian astronomers in the third or fourth century, using a pre-Ptolemaic Greek table of chords, produced tables of sines and versines, from which it was trivial to derive cosines. This new system of trigonometry, produced in India, was transmitted to the Arabs in the late eighth century and by them, in an expanded form, to the Latin West and the Byzantine East in the twelfth century."
9. (Bourbaki 1998, p. 126): "As for trigonometry, it is disdained by geometers and abandoned to surveyors and astronomers; it is these latter (Aristarchus, Hipparchus, Ptolemy) who establish the fundamental relations between the sides and angles of a right angled triangle (plane or spherical) and draw up the first tables (they consist of tables giving the *chord* of the arc cut out by an angle $\theta < \pi$ on a circle of radius r , in other words the number $2r \sin(\theta/2)$; the introduction of the sine, more easily handled, is due to Hindu mathematicians of the Middle Ages)."

10. (Filliozat 2004, pp. 140–143)
11. (Hayashi 1995)
12. (Kim Plofker 2007, p. 6)
13. (Stillwell 2004, p. 173)
14. (Bressoud 2002, p. 12) Quote: "There is no evidence that the Indian work on series was known beyond India, or even outside Kerala, until the nineteenth century. Gold and Pingree assert [4] that by the time these series were rediscovered in Europe, they had, for all practical purposes, been lost to India. The expansions of the sine, cosine, and arc tangent had been passed down through several generations of disciples, but they remained sterile observations for which no one could find much use."
15. (Plofker 2001, p. 293) Quote: "It is not unusual to encounter in discussions of Indian mathematics such assertions as that "the concept of differentiation was understood [in India] from the time of Manjula (... in the 10th century)" [Joseph 1991, 300], or that "we may consider Madhava to have been the founder of mathematical analysis" (Joseph 1991, 293), or that Bhaskara II may claim to be "the precursor of Newton and Leibniz in the discovery of the principle of the differential calculus" (Bag 1979, 294). ... The points of resemblance, particularly between early European calculus and the Keralese work on power series, have even inspired suggestions of a possible transmission of mathematical ideas from the Malabar coast in or after the 15th century to the Latin scholarly world (e.g., in (Bag 1979, 285)). ... It should be borne in mind, however, that such an emphasis on the similarity of Sanskrit (or Malayalam) and Latin mathematics risks diminishing our ability fully to see and comprehend the former. To speak of the Indian "discovery of the principle of the differential calculus" somewhat obscures the fact that Indian techniques for expressing changes in the Sine by means of the Cosine or vice versa, as in the examples we have seen, remained within that specific trigonometric context. The differential "principle" was not generalised to arbitrary functions—in fact, the explicit notion of an arbitrary function, not to mention that of its derivative or an algorithm for taking the derivative, is irrelevant here"

16. (Pingree 1992, p. 562) Quote:"One example I can give you relates to the Indian Mādhava's demonstration, in about 1400 A.D., of the infinite power series of trigonometrical functions using geometrical and algebraic arguments. When this was first described in English by Charles Matthew Whish, in the 1830s, it was heralded as the Indians' discovery of the calculus. This claim and Mādhava's achievements were ignored by Western historians, presumably at first because they could not admit that an Indian discovered the calculus, but later because no one read anymore the *Transactions of the Royal Asiatic Society*, in which Whish's article was published. The matter resurfaced in the 1950s, and now we have the Sanskrit texts properly edited, and we understand the clever way that Mādhava derived the series *without* the calculus; but many historians still find it impossible to conceive of the problem and its solution in terms of anything other than the calculus and proclaim that the calculus is what Mādhava found. In this case the elegance and brilliance of Mādhava's mathematics are being distorted as they are buried under the current mathematical solution to a problem to which he discovered an alternate and powerful solution."
17. (Katz 1995, pp. 173–174) Quote:"How close did Islamic and Indian scholars come to inventing the calculus? Islamic scholars nearly developed a general formula for finding integrals of polynomials by A.D. 1000—and evidently could find such a formula for any polynomial in which they were interested. But, it appears, they were not interested in any polynomial of degree higher than four, at least in any of the material that has come down to us. Indian scholars, on the other hand, were by 1600 able to use ibn al-Haytham's sum formula for arbitrary integral powers in calculating power series for the functions in which they were interested. By the same time, they also knew how to calculate the differentials of these functions. So some of the basic ideas of calculus were known in Egypt and India many centuries before Newton. It does not appear, however, that either Islamic or Indian mathematicians saw the necessity of connecting some of the disparate ideas that we include under the name calculus. They were apparently only interested in specific cases in which these ideas were needed. ... There is no danger, therefore, that we will have to rewrite the history texts to remove the statement that Newton and Leibniz invented calculus. They were certainly the ones who were able to combine many differing ideas under the two unifying themes of the derivative and the integral, show the connection between them, and turn the calculus into the great problem-solving tool we have today."
18. Sergent, Bernard (1997), *Genèse de l'Inde* (in French), Paris: Payot, p. 113, ISBN 978-2-228-89116-5

19. Coppa, A.; et al. (6 April 2006), "Early Neolithic tradition of dentistry: Flint tips were surprisingly effective for drilling tooth enamel in a prehistoric population", *Nature*, **440** (7085): 755–6, Bibcode:2006Natur.440..755C (<https://ui.adsabs.harvard.edu/abs/2006Natur.440..755C>) , doi:10.1038/440755a (<https://doi.org/10.1038%2F440755a>) , PMID 16598247 (<https://pubmed.ncbi.nlm.nih.gov/16598247/>) , S2CID 6787162 (<https://api.semanticscholar.org/CorpusID:6787162>) .
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22. A. Seidenberg, 1978. The origin of mathematics. Archive for History of Exact Sciences, vol 18.
23. (Staal 1999)
24. (Hayashi 2003, p. 118)
25. (Hayashi 2005, p. 363)
26. Pythagorean triples are triples of integers (a, b, c) with the property: $a^2+b^2 = c^2$. Thus, $3^2+4^2 = 5^2$, $8^2+15^2 = 17^2$, $12^2+35^2 = 37^2$, etc.
27. (Cooke 2005, p. 198): "The arithmetic content of the *Śulva Sūtras* consists of rules for finding Pythagorean triples such as (3, 4, 5), (5, 12, 13), (8, 15, 17), and (12, 35, 37). It is not certain what practical use these arithmetic rules had. The best conjecture is that they were part of religious ritual. A Hindu home was required to have three fires burning at three different altars. The three altars were to be of different shapes, but all three were to have the same area. These conditions led to certain "Diophantine" problems, a particular case of which is the generation of Pythagorean triples, so as to make one square integer equal to the sum of two others."

28. (Cooke 2005, pp. 199–200): "The requirement of three altars of equal areas but different shapes would explain the interest in transformation of areas. Among other transformation of area problems the Hindus considered in particular the problem of squaring the circle. The *Bodhayana Sutra* states the converse problem of constructing a circle equal to a given square. The following approximate construction is given as the solution.... this result is only approximate. The authors, however, made no distinction between the two results. In terms that we can appreciate, this construction gives a value for π of $18(3 - 2\sqrt{2})$, which is about 3.088."
29. (Joseph 2000, p. 229)
30. "Vedic Maths Complete Detail" (<https://www.theintellibrain.com/vedicmaths/>) . ALLEN IntelliBrain. Retrieved 22 October 2022.
31. (Cooke 2005, p. 200)
32. The value of this approximation, 577/408, is the seventh in a sequence of increasingly accurate approximations $3/2, 7/5, 17/12, \dots$ to $\sqrt{2}$, the numerators and denominators of which were known as "side and diameter numbers" to the ancient Greeks, and in modern mathematics are called the Pell numbers. If x/y is one term in this sequence of approximations, the next is $(x + 2y)/(x + y)$. These approximations may also be derived by truncating the continued fraction representation of $\sqrt{2}$.
33. Neugebauer, O. and A. Sachs. 1945. *Mathematical Cuneiform Texts*, New Haven, CT, Yale University Press. p. 45.
34. Mathematics Department, University of British Columbia, *The Babylonian tabled Plimpton 322* (<http://www.math.ubc.ca/~cass/courses/m446-03/pl322/pl322.html>) Archived (<https://web.archive.org/web/20200617151320/http://www.math.ubc.ca/~cass/courses/m446-03/pl322/pl322.html>) 17 June 2020 at the Wayback Machine.
35. Three positive integers (a, b, c) form a *primitive* Pythagorean triple if $c^2 = a^2 + b^2$ and if the highest common factor of a, b, c is 1. In the particular Plimpton322 example, this means that $13500^2 + 12709^2 = 18541^2$ and that the three numbers do not have any common factors. However some scholars have disputed the Pythagorean interpretation of this tablet; see Plimpton 322 for details.
36. (Dani 2003)

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38. (Fowler 1996, p. 11)
39. (Singh 1936, pp. 623–624)
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42. (Filliozat 2004, p. 137)
43. (Pingree 1988, p. 637)

44. (Staal 1986)
45. (Filliozat 2004, p. 139)
46. (Filliozat 2004, pp. 140–141)
47. (Yano 2006, p. 146)
48. (Filliozat 2004, pp. 143–144)
49. (Filliozat 2004, p. 144)
50. (Pingree 1988, p. 638)
51. (Hayashi 2003, pp. 122–123)
52. (Hayashi 2003, p. 123)
53. (Hayashi 2003, p. 119)
54. (Plofker 2007, p. 395)
55. (Plofker 2007, p. 395); (Plofker 2009, pp. 47–48)
56. (Hayashi 2005, p. 366)
57. (Plofker 2009, p. 45)
58. (Plofker 2009, p. 46)
59. (Plofker 2009, p. 47)
60. (Plofker 2009)
61. (Pingree 1978, p. 494)
62. (Plofker 2009, p. 48)
63. (Hayashi 2005, p. 371)
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71. (Katz 1995)
72. (Hayashi 2005, p. 369)
73. (Hayashi 2003, pp. 121–122)
74. (Stillwell 2004, p. 77)
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Further reading

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- Mathematics in ancient India by R. Sridharan (<https://drive.google.com/file/d/0BxearzN4q-babEVJZ2g0VnZCQzA/view?usp=sharing>)
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